

Multiple integrals – 5

Asymptotics of multiple integrals

Homework

13 сентября 2026 года

Theoretical minimum. When a large parameter enters an integral through an exponent, the mass concentrates where the exponent is maximal, and the answer is the value there times the Gaussian volume of the neighbourhood: this is Laplace's method, with the factor $(2\pi/\lambda)^{n/2}/\sqrt{|\det S''|}$ in dimension n . If the maximum sits on the boundary the exponent of λ changes, which is why integrals like $n^2 \iint x^n y^n f$ behave as they do — there the mass collects at a corner. A family of non-negative kernels of unit mass concentrating at a point reproduces the value of a continuous function, and the rate is governed by the modulus of continuity. On the sphere and the ball in high dimension the measure itself concentrates, so averaging a Lipschitz function is the same as evaluating it at a typical point. Two practical rules: first localise — replace the domain by a shrinking neighbourhood of the maximum — and only then expand the exponent to second order.

1. Prove Watson's lemma in the multidimensional form: for f continuous near 0 with $f(0) \neq 0$ the integral $\int_{\|x\| \leq \delta} f(x) e^{-\lambda \|x\|^2} dx$ is asymptotically $f(0) \left(\frac{\pi}{\lambda}\right)^{n/2}$, and estimate the error for Hölder continuous f .
2. Find the asymptotics of $\iint_{[0,1]^2} (xy)^n dx dy$, of $\iint_{[0,1]^2} (x+y)^n dx dy$ and of $\iint_{x^2+y^2 \leq 1} (1-x^2-y^2)^n dx dy$, and say in each case where the mass concentrates.
3. Prove that $n^2 \iint_{[0,1]^2} x^n y^n f(x, y) dx dy \rightarrow f(1, 1)$ for every continuous f , and give the rate for f Lipschitz; compare with problem 2 of the class sheet, where the next term is computed.

4. Find the asymptotics of the volume of the intersection of the cube $[-1, 1]^n$ with the ball of radius $\sqrt{n}/2$, and explain the answer through the central limit theorem (Probability – 3).
5. Prove that $\int_{S^{n-1}} e^{\lambda x_1} d\sigma(x)$ is asymptotically $\sigma(S^{n-1}) \frac{e^\lambda}{\lambda^{(n-1)/2}} c_n$ for a constant c_n to be found, and interpret the result as concentration near the pole.
6. Let f be continuous on $\mathbb{R}^n$ with compact support. Prove that $\left(\frac{n}{\pi}\right)^{n/2} \int_{\mathbb{R}^n} e^{-n\|x-a\|^2} f(x) dx \rightarrow f(a)$ and find the first correction term for $f \in C^2$.
7. Find $\lim_{n \rightarrow \infty} n \iiint_{x^2+y^2+z^2 \leq 1} (1 - \sqrt{x^2 + y^2 + z^2})^n f(x, y, z) dx dy dz$ for continuous f , after normalising the answer correctly.
8. Prove the multidimensional stationary phase estimate $|\int e^{i\lambda S(x)} f(x) dx| = O(\lambda^{-n/2})$ for a non-degenerate critical point, and explain why the one-dimensional case of Improper integrals – 2 gives $\lambda^{-1/2}$.
9. Let A be a symmetric positive definite matrix. Find the asymptotics of $\int_{\mathbb{R}^n} e^{-\lambda \langle Ax, x \rangle} \|x\|^2 dx$ as $\lambda \rightarrow \infty$ and as $n \rightarrow \infty$ with λ fixed, and compare the two regimes.
10. Prove that $\frac{1}{\text{vol } B_n} \int_{B_n} f(\langle x, e \rangle \sqrt{n}) dx \rightarrow \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} f(t) e^{-t^2/2} dt$ for bounded continuous f and a fixed unit vector e , and say which statement of Multiple integrals – 3 this is a restatement of.